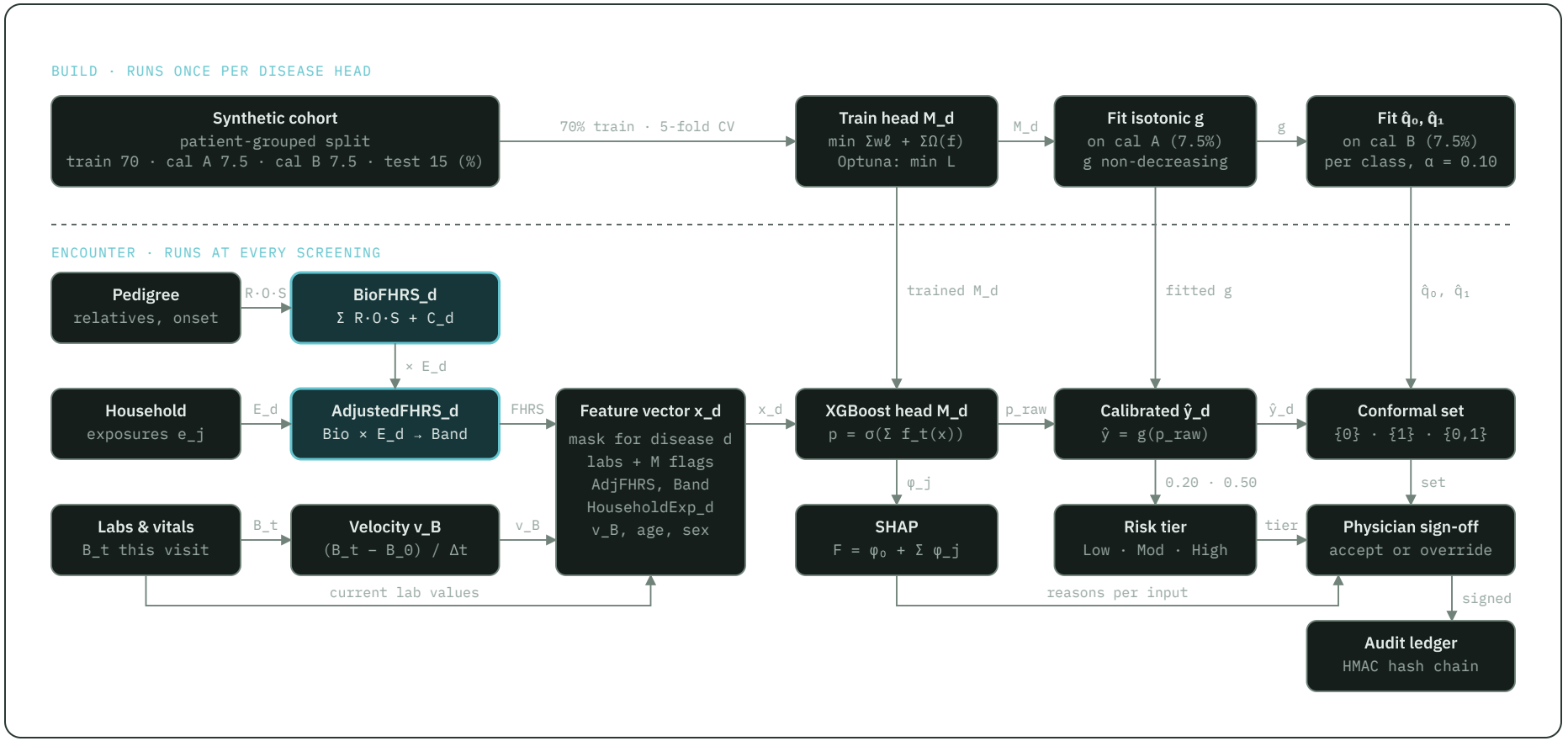
Every formula the system uses, in the order it runs. A single synthetic patient is followed from family history to a signed risk tier.



The build lane fixes three things once per disease head: the trained XGBoost model, the isotonic calibrator g, and the conformal thresholds. Every screening then reuses them. The highlighted boxes are the Family Health Risk Score engine. The 15% test partition enters neither lane: it is scored once for the

final report.

EXAMPLE PATIENT

P-0417, screened for type 2 diabetes

RECORD	AGE · SEX	HEAD	ENCOUNTER	HORIZON
Synthetic cohort	46 · female	T2D	Visit 2, 4 months after baseline	5-year onset

EXACT Steps E1–E4 are exact arithmetic from the formulas. ILLUSTRATIVE Steps E5–E9 depend on a trained model, so their numbers are placeholders. Replace them with your trained T2D head's outputs.

BUILD LANE

Before any patient is seen

These three fits happen once per disease head, on data that never overlaps.

B1 · training partition, 70%

Train and tune the head

$$\text{Obj} = \Sigma_i\, w_i \cdot \ell(y_i, p_i) + \Sigma_t\, \Omega(f_t)$$

$$\ell = -[\, y \ln p + (1 - y) \ln(1 - p) \,]$$

$$\Omega(f) = \gamma T + \tfrac{1}{2}\lambda \|w\|^2$$

$$w_i = \text{scale_pos_weight if } y_i = 1, \text{ else } 1$$

ℓ log loss; Ω penalizes trees with many leaves (T) or large leaf values (w).

scale_pos_weight makes a missed sick patient cost more; it

B2 · calibration half A, 7.5%

Fit the calibrator g

$$g = \text{argmin}\, \Sigma_i\, (\, y_i - g(p_i)\,)^2$$

$$\text{subject to } g \text{ non-decreasing}$$

The trained head scores every patient in half A. Isotonic regression then finds the step function that best matches those scores to observed outcomes without changing their order. With fewer than about 1,000 samples, a sigmoid fit replaces it.

`scale_pos_weight` makes a missed sick patient cost more; it starts at negatives ÷ positives.

$$L_{Optuna} = 0.40(1 - AUROC) + 0.20(1 - AUPRC) + 0.20\ ECE + 0.10\ Brier + 0.10\ \Delta AUROC_{subgroup}$$

Optuna picks the hyperparameters (tree depth, learning rate, γ, λ, scale_pos_weight) that minimize L averaged over five patient-grouped folds.

B3 · calibration half B, 7.5%

Fit the conformal thresholds

$$s_i = 1 - \hat{p}_{y_i}(x_i)$$

$$\hat{q}_y = \lceil (n_y + 1)(1 - \alpha) \rceil\text{-th smallest } s_i \text{ among half-B patients with label } y$$

Each patient's score is one minus the calibrated probability of their true label. Computing a threshold separately for each class (Mondrian) keeps 90% coverage for the rare positive class too. Here $\hat{p}_1 = \hat{y}$ and $\hat{p}_0 = 1 - \hat{y}$.

ENCOUNTER LANE

One screening, step by step

Formulas on the left; the example patient's numbers on the right.

E1 · FHRS ENGINE

EXAMPLE · T2D

EXACT

Family history → $BioFHRS_d$

$$BioFHRS_d = \Sigma_i\ R_i \cdot O_i \cdot S_i + C_d$$

$$C_d = \Sigma_{k=1..3}\ W_k \cdot \max(0,\ N_k^{ver} - 1)$$

R kinship: 0.50 first-degree · 0.25 second · 0.125 third

O onset: 3.00 severe early · 2.00 early · 1.00 typical · 0.75 late (disease-specific ages)

S 1.0 confirmed by records · 0.5 self-reported

W 1.0 · 0.5 · 0.25 by tier; N^{ver} verified relatives in that tier

RELATIVE	R	O	S	R·O·S
Father, T2D at 35, records	0.50	2.00	1.0	1.00
Mother, T2D at 55, self-report	0.50	1.00	0.5	0.25
Paternal grandmother, T2D at 60, records	0.25	1.00	1.0	0.25
Clustering C_{T2D}: 1 verified in tier 1, 1 in tier 2				0.00

$BioFHRS_{T2D} = 1.00 + 0.25 + 0.25 + 0$ **1.50**

Onset 35 falls in the T2D Early band (30–39), so O = 2.00. The mother is unverified, so she does not count toward clustering.

E2 · FHRS ENGINE

Household → $AdjustedFHRS_d$ → Band

$$E_d = 1.0 + \Sigma_j\ e_{j,d}, \quad e_{j,d} \in [0.05,\ 0.35]$$

$$AdjustedFHRS_d = BioFHRS_d \times E_d$$

Band: 0 = 0 · 1 (0, 1.00] · 2 (1.00, 2.50] 3 (2.50, 5.00] · 4 > 5.00

E_d is also passed to the model on its own as `HouseholdExposure_d`, so exposure still counts when $BioFHRS_d$ = 0.

EXAMPLE · T2D — EXACT

Household diet high in refined carbohydrates	e = 0.15
$E_{T2D} = 1.0 + 0.15$	1.15
$AdjustedFHRS_{T2D} = 1.50 \times 1.15$	1.725

1.725 falls in (1.00, 2.50] **Band 2 · Moderate**

The weight 0.15 is an example; use your calibrated $e_{j,d}$ values.

E3 · LONGITUDINAL TRIAGE — EXAMPLE · AT = 4 MONTHS — EXACT

E3 · LONGITUDINAL TRIAGE

Labs → biomarker velocity

$$\Delta B = B_t - B_0$$

$$v_B = \Delta B / \Delta t \quad (\Delta t \text{ in months})$$

Velocity is NaN, with its missingness flag set to 1, at the first visit and whenever Δt is under 30 days.

EXAMPLE · Δt = 4 MONTHS — EXACT

BIOMARKER	B_0	B_T	ΔB	V_B / MONTH
Fasting glucose, mg/dL	92	118	+26	6.5
HbA1c, %	5.4	5.9	+0.5	0.125

Glucose 118 mg/dL is still below the 126 mg/dL diabetes cutoff; the velocity carries the warning.

E4 · FEATURE ROUTING

Assemble the feature vector x_d

$$x_d = \text{mask}_d(\text{ labs, M flags, AdjustedFHRS}_d,$$
$$\text{Band}_d, \text{HouseholdExposure}_d, v_B, \text{age, sex })$$
$$M = 1 \text{ if the value is missing, else } 0$$

The mask keeps only the features relevant to disease d. Missing values stay NaN; their flag records that the test was not done.

EXAMPLE · X_{T2D} — EXACT

Age · sex	46 · F
Fasting glucose · HbA1c	118 mg/dL · 5.9 %
Fasting insulin · $M_{insulin}$	NaN · 1
BMI · waist-to-hip ratio	29.1 kg/m² · 0.88
AdjustedFHRS · Band	1.725 · 2
$\text{HouseholdExposure}_{T2D}$	1.15
$v_{glucose} \cdot v_{HbA1c}$	6.5 · 0.125

BMI and waist-to-hip ratio are example values for this patient.

E5 · MODEL

XGBoost head → raw score

$$F(x) = \Sigma_{t=1..T}\, f_t(x)$$

$$p_{raw} = 1 / (1 + e^{-F(x)})$$

EXAMPLE — ILLUSTRATIVE

Sum of leaf values F(x)	0.45
$p_{raw} = 1 / (1 + e^{-0.45})$	0.61

Each tree f_t sends x down its splits to a leaf value; a NaN follows the branch learned during training. The leaf values add up to the log-odds F.

This is high because scale_pos_weight inflated the training loss for positives. Step E7 corrects it.

E6 · EXPLANATION

SHAP → reasons per input

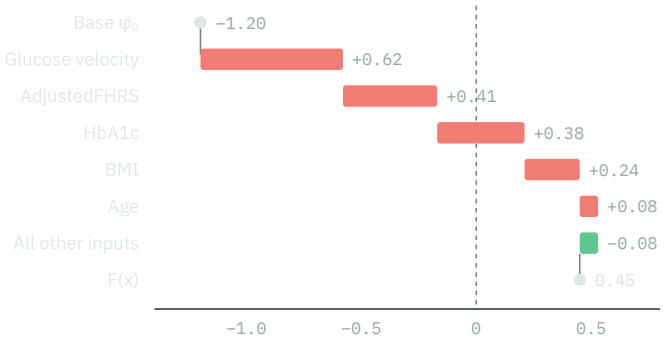
$$F(x) = \varphi_0 + \Sigma_j \varphi_j$$

$$I_j = (1/N)\ \Sigma_i\ |\varphi_{ij}| \quad \text{(global, Chapter 4)}$$

φ_0 is the cohort's average log-odds; each φ_j is how much input j moved this patient's score. The contributions add up exactly to F(x). Averaging |φ| over all patients ranks inputs by overall importance.

EXAMPLE · LOG-ODDS

ILLUSTRATIVE



Red bars raise risk, green bars lower it. These values come from the raw model score, before calibration.

E7 · CALIBRATION

Raw score → calibrated risk $\hat{y}_d$

$$\hat{y}_d = g(p_{raw})$$

g is the step function fitted in B2. It turns the inflated raw score into an honest rate: among patients given ŷ = 0.34, about 34 in 100 are diagnosed within 5 years.

EXAMPLE

ILLUSTRATIVE

p_{raw}	0.61
$\hat{y}_{T2D} = g(0.61)$	0.34

5-year T2D risk **34%**

E8 · UNCERTAINTY + E9 · TIER

Conformal set and risk tier

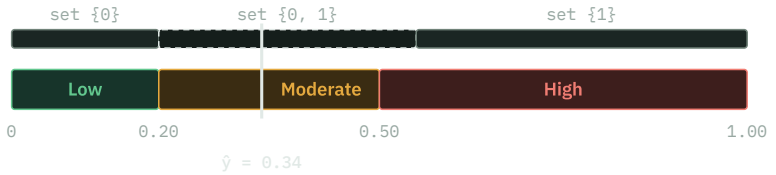
$$1 \in C(x) \text{ if } 1 - \hat{y} \le \hat{q}_1$$

$$0 \in C(x) \text{ if } \hat{y} \le \hat{q}_0$$

Low $\hat{y} < 0.20$ · Moderate $0.20 \le \hat{y} < 0.50$ · High $\hat{y} \ge$

C(x) contains the true outcome for at least 90% of patients. A set of {0, 1} means the model cannot rule out either outcome, so the tier is shown with an "Uncertain" flag.

EXAMPLE · $\hat{Q}_1 = 0.80$, $\hat{Q}_0 = 0.55$ ILLUSTRATIVE



$1 - 0.34 = 0.66 \le 0.80$ and $0.34 \le 0.55$

Moderate Uncertain {0, 1}

Guidance for this tier: lifestyle intervention, follow-up within 3–6 months, confirmatory panel.

E10 · GOVERNANCE

Physician sign-off → audit ledger

$$\text{Digest}_n = \text{HMAC-SHA-256}(\ K,\ \text{PseudoID} \parallel \text{Timestamp} \parallel X \parallel \text{FHRS} \parallel \hat{y} \parallel \text{PhysicianSignature} \parallel \text{Digest}_{n-1}\ )$$

Each entry includes the previous entry's digest, so changing any past record breaks every digest after it. The key K is held outside the database.

EXAMPLE ILLUSTRATIVE

Physician decision	Accepts · orders OGTT
Sealed fields	x_{T2D}, 1.725, 0.34, signature
Chained to	Digest of visit 1

The physician's decision, not the model's tier, is what goes on record.